

UBUNTU SERVER 22.04 LTS INSTALLATION & DEPLOYMENT MANUAL

Standard Operating Procedure for Enterprise Virtual Environments

Prepared By: Mar Aldrei Y. Avila

Role: Junior System Administrator / Student

Course: System Administration and Maintenance

Section: BSIT 4A

Target Audience: Systems & Network Administrators

Date: August 15, 2026

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1. INTRODUCTION

This manual provides an end-to-end, reproducible guide for deploying, configuring, and verifying an Ubuntu Server 22.04 LTS virtual machine instance. Designed for system administrators, this document outlines standard operating procedures to establish a hardened, production-ready Linux server environment.

By following these standardized instructions, administrators can ensure consistent resource provisioning, proper network initialization, and reliable hypervisor integration across enterprise sandboxes or local server infrastructure.

2. SYSTEM & VIRTUAL MACHINE REQUIREMENTS

Before initiating deployment, verify that the host hypervisor meets or exceeds the baseline virtual resource specifications outlined below:

Resource / Parameter	Minimum Specification	Recommended Deployment Spec
Host Hypervisor	Oracle VM VirtualBox 7.0+	VirtualBox 7.0+ / VMware ESXi / Workstation
Processor (vCPU)	1 Core	2 Execution Cores (VT-x/AMD-V Enabled)
Base Memory (RAM)	2048 MB (2 GB)	4096 MB (4 GB)
Virtual Disk Storage	15.00 GB VDI	25.00 GB Dynamically Allocated VDI
Network Adapter	NAT Mode	Bridged Adapter / Dual NIC (NAT + Host-Only)

3. UBUNTU SERVER INSTALLATION GUIDE

Follow these sequential instructions to deploy the base operating system image:

Step 1: Hypervisor Provisioning

1. Launch VirtualBox and click **New** to open the Virtual Machine Creation Wizard.
2. Set Name to `Ubuntu-Server-22.04`, Type to `Linux`, and Version to `Ubuntu (64-bit)`.
3. Allocate **4096 MB** RAM and assign **2 Processor Cores**.
4. Select **Create a Virtual Hard Disk Now**, choose **VDI (VirtualBox Disk Image)**, set storage to **Dynamically Allocated**, and assign **25.00 GB** capacity.

Step 2: ISO Attachment & First Boot

1. Navigate to `VM Settings > Storage`.

2. Click the optical drive icon under Storage Devices, choose **Choose a disk file...**, and attach the downloaded `ubuntu-22.04.x-live-server-amd64.iso` image.
3. Start the VM and select **Try or Install Ubuntu Server** from the GRUB boot menu.

Step 3: Installer Configuration

1. Select primary language (English) and apply installer updates if prompted.
2. Choose default keyboard layout (English US).
3. Choose base installation profile: Select standard **Ubuntu Server**.
4. Configure Network Interface: Accept automatic DHCP configuration or assign static IP parameters.
5. Storage Guided Setup: Select **Use an entire disk** and ensure *Set up this disk as an LVM group* is selected for flexible storage management.
6. User Account Profile: Input primary administrative account details:
 - **Your Name:** Admin User
 - **Server Name:** primary-server-01
 - **Username:** sysadmin
 - **Password:** Configured administrative password
7. Enable **Install OpenSSH Server** to allow remote management access.
8. Complete installation, unmount the installation ISO, and reboot the virtual machine.

Figure 1: Ubuntu Server Guided Storage & User Profile Setup.

4. INITIAL SERVER CONFIGURATION

Once the server boots up to the login shell, execute the following foundational configuration tasks:

1. System Package Updates

Update repository indexes and upgrade system packages to current security releases:

```
sudo apt update && sudo apt upgrade -y
```

2. Firewall & Remote Access Hardening

Enable Uncomplicated Firewall (UFW) and explicitly allow SSH connections:

```
sudo ufw allow ssh
sudo ufw enable
sudo ufw status verbose
```

3. Guest Tools & Integration Installation

Install hypervisor guest utilities for operational monitoring and memory driver optimizations:

```
sudo apt install -y open-vm-tools virtualbox-guest-utils
```

5. SERVER VERIFICATION PROCEDURES

To confirm that the server deployment meets quality standards, execute and document the following system audit commands:

```
# 1. Audit Active Kernel and OS Version
uname -a

# 2. Inspect Storage Volumes and Partition Layout
lsblk -f

# 3. Verify Memory Allocation and Usage Status
free -h
```

```
# 4. Check Network Interfaces and IP Addresses
ip a

# 5. Confirm Boot Mode Architecture (UEFI vs BIOS)
[ -d /sys/firmware/efi ] && echo "Firmware: UEFI Mode" || echo "Firmware:
Legacy BIOS Mode"
```

Figure 2: CLI Verification Output Demonstrating Active Kernel and Storage Layout.

6. TROUBLESHOOTING GUIDE

Below are common issues encountered during virtual machine deployment along with their definitive technical solutions:

Observed Issue	Probable Root Cause	Remediation / Solution
VT-x / AMD-V Error on Launch	Hardware Virtualization disabled in host motherboard firmware.	Reboot host PC, enter Host BIOS/UEFI settings, enable <i>Intel Virtualization Technology</i> or <i>AMD-V</i> , save and reboot.
No Network / IP Address Assigned	Hypervisor virtual NIC misconfigured or DHCP server unresponsive.	Verify VM Network settings are set to NAT or Bridged mode. Restart networking service: <code>sudo systemctl restart systemd-networkd</code> .
Installer Freezes / Black Screen	Graphics driver mismatch or corrupted ISO file download.	Set VM Graphics Controller to <i>VMSVGA</i> , check display 3D acceleration, and verify ISO MD5/SHA256 checksums.
Kernel Panic on Boot	Insufficient memory allocation or corrupted bootloader files.	Increase allocated vRAM to at least 2048 MB and re-verify EFI partition configuration during disk partitioning.

7. SYSTEM ADMINISTRATOR BEST PRACTICES

- **Implement Least Privilege Access:** Never perform daily administrative tasks directly as the root account. Use a dedicated user account with `sudo` privileges.
- **Enforce SSH Key-Based Authentication:** Disable password-based SSH authentication in `/etc/ssh/sshd_config` and utilize 4096-bit RSA or Ed25519 SSH key pairs.
- **Establish Virtual Machine Snapshots:** Always create a VirtualBox snapshot immediately following base OS configuration and verification. This provides a baseline rollback point prior to installing major roles or applications.
- **Automate Package Security Patching:** Install and configure `unattended-upgrades` to ensure critical security patches are automatically applied:

```
sudo apt install unattended-upgrades
```

- **Standardize Disk Partition Layouts:** Use Logical Volume Management (LVM) for root partitions to allow online volume extension when storage capacity needs grow.